Lesson notes · for you

Secret messages: encryption basics

Cyber security · Level 3 · Lesson 4

Time	Materials
20 min	Paper, a pen and scissors; Two strips of paper with the alphabet written on each

Learning goals

- Explain encryption, keys and decryption.
- Use Caesar's cipher, and crack it by trying keys or counting letters.
- Explain why modern encryption is strong, and where it protects you.

Activity: Cipher challenge

1. Make (4 min): Write the alphabet on two strips of paper. Slide one along the other to make any shift, and mark where A (or ا) lines up.
2. Hide (5 min): Choose a secret shift and encrypt a message of at least three sentences. Longer messages are easier to crack by counting, so try both a short and a long one.
3. Crack (8 min): Swap messages with a partner. First try shifts one by one; then count the letters and guess that the most common one stands for E, or alif.
4. Compare (3 min): Which method found the key faster? Why would neither work on the encryption in your chat app?

Arabic messages: use the 28 letters in order (ا ب ت ث ...), and write آ إ أ as ا, ؤ as و, ئ and ى as ي, and ة as ه, as the Labs app does.

Discussion

- Why do long messages make Caesar's cipher easier to crack?
- If the method is public, what keeps a message secret?
- Why do attackers trick people instead of breaking encryption?

Check yourself

- One place encryption protects me + one key I'll keep to myself.

Try it this week

Write a short message to a friend in Caesar's cipher and tell them the key in person. Then swap: each of you tries to crack the other's next message without the key, by trying shifts or counting letters. Time how long it takes.

Worksheet

Secret messages: encryption basics

Cyber security · Level 3 · Lesson 4

Name _____ Class _____ Date _____

1 **Why is Caesar's cipher easy to crack?**

◯ Because it's very old ◯ There are so few keys you can try them all

2 **Decrypt this message. It was written with Caesar's cipher and a shift of 3: KHOOR**

_____

3 **End-to-end encryption means the app company can read your messages, but hackers can't.**

◯ True ◯ False

4 **Match each word to its meaning.**

Plaintext	• • • •	Turning ciphertext back into plaintext
Ciphertext	• • • •	The secret that locks and unlocks it
Key	• • • •	The scrambled message
Decryption	• • • •	The readable message

5 **In which of these does encryption protect what you share?**

☐ A WhatsApp chat between you and a friend ☐ A page you open over HTTPS, with the padlock

☐ Photos on a phone with a screen lock ☐ Your message, after a friend forwards it to a group

☐ A password you told a classmate

✎ **Where does encryption already protect you every day? And where might you be giving away a key without thinking?**

_____

_____

_____